

WHY DO WE FALL ILL

Class 9 | Science | Chapter 13

HEALTH & DISEASE

Health = a state of complete physical, mental & social well-being -- not merely the absence of disease.

Disease = a condition in which the normal functioning of the body is disturbed.

Health of an individual also depends on the community/surroundings -- public hygiene, clean water, and social harmony matter.

ACUTE VS CHRONIC DISEASE

Acute Disease	Chronic Disease
Lasts for a short time	Lasts for a long time (even a lifetime)
Usually no lasting effect on health	Can have a bigger, long-term effect on health

Ex: Acute -- common cold. Chronic -- elephantiasis, tuberculosis.

CAUSES OF DISEASE

Infectious & Non-Infectious

CAUSES OF DISEASE

Type	Meaning	Examples
Infectious disease	Caused by a biological agent (pathogen/microbe) that CAN spread from person to person	Cold, TB, Cholera, AIDS
Non-infectious disease	NOT caused by an external agent that spreads; e.g. genetic, lifestyle related	Cancer, high blood pressure, allergies

TYPES OF INFECTIOUS AGENTS

Agent	Disease Example
Virus	Common cold, AIDS, influenza
Bacteria	Typhoid, tuberculosis, cholera
Fungi	Skin infections (ringworm)
Protozoa	Malaria (Plasmodium), Kala-azar
Worms	Elephantiasis, intestinal worm infections

MEANS OF SPREAD

Mode	Example
Air	Common cold, TB (via coughing/sneezing)
Water	Cholera, typhoid
Vector-borne	Malaria & dengue via mosquitoes
Sexual contact/blood	HIV-AIDS

PRINCIPLES OF TREATMENT & PREVENTION

Immunisation & Public Health

PRINCIPLES OF TREATMENT

- Reduce the effect of disease (symptomatic treatment, e.g. medicine for fever/pain).
- Kill the cause of disease -- antibiotics for bacteria; antibiotics do NOT work on viruses.

Antibiotics block bacterial processes (like cell-wall making) that human cells don't have, so they don't harm us -- but they are ineffective against viruses.

PRINCIPLES OF PREVENTION

- General prevention: clean water, sanitation, nutrition, reduced overcrowding -- reduces exposure to ALL infectious diseases.
- Specific prevention: immunisation against a particular disease.

IMMUNISATION

The process of making the body immune/resistant to a specific infection, usually via a vaccine.

Principle: the immune system 'remembers' a pathogen it has met before and responds quickly on re-exposure. Ex: vaccines for polio, measles, hepatitis, DPT.

SUMMARY & REVISION

Key Terms + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Health** -> complete physical, mental & social well-being
- **Acute disease** -> short duration; **Chronic disease** -> long duration
- **Infectious disease** -> caused by a spreadable pathogen
- **Antibiotics** -> kill/block bacteria, ineffective against viruses
- **Immunisation** -> makes the body resistant to a specific disease

SIGNS & SYMPTOMS

Symptoms = what the patient feels/reports (e.g. headache). Signs = what a doctor can observe/measure (e.g. fever, swelling).

ONE-LINE RECAP FOR REVISION

Health depends on individual habits AND the community/environment.

Infectious diseases spread; non-infectious ones generally don't.

Prevention (hygiene + immunisation) is better than only treating symptoms.

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